

TR-CORE-001

- Related Functional Requirement: FR-1
- Objective: Implement reliable source ingestion of customer and order CSVs into Delta tables.
- Target Cluster Configuration:
 - Cluster Name: ingestion-cluster
 - Runtime Version: 11.3.x (Databricks Runtime)
 - Node Type: m5.xlarge
 - Driver Node: 1 (4 vCPU, 16GB RAM)
 - Worker Nodes: 2 (4 vCPU, 16GB RAM)
 - Autoscaling: Enabled (2-8 nodes)
 - Auto Termination: 30 minutes
 - Libraries: delta, pyspark
- Source Data Details:
 - Source Type: CSV
 - Location: /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata, /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata
 - Schema: customer (CustId, Name, EmailId, Region); order (OrderId, ItemName, PricePerUnit, Qty, Date, CustId)
- Target Data Details:
 - Table Name: customer, order
 - Storage Format: Delta
 - Partitioning Strategy: None specified
- Job Flow:
 1. Read source CSV files
 2. Infer or apply specified schema
 3. Write as Delta tables
- Transformations & Logic:
 - Basic type and schema validation
- Error Handling & Logging:
 - Validate source file presence
 - Log ingestion errors and write failures

TR-CORE-002

- Related Functional Requirement: FR-2
- Objective: Enforce explicit schemas for ingested customer and order tables.
- Target Cluster Configuration:
 - Cluster Name: schema-enforcement-cluster
 - Runtime Version: 11.3.x
 - Node Type: m5.xlarge
 - Driver Node: 1
 - Worker Nodes: 2
 - Autoscaling: Enabled (2-8)
 - Auto Termination: 30 minutes
 - Libraries: pyspark
- Source Data Details:
 - Source Type: Delta
 - Location: as written above
 - Schema: as per requirements
- Target Data Details:
 - Table Name: customer, order
 - Storage Format: Delta
- Job Flow:
 1. Apply schema during load or alter table to enforce
 2. Reject non-conforming records
- Transformations & Logic:
 - Schema enforcement
- Error Handling & Logging:
 - Log schema mismatches

TR-CORE-003

- Related Functional Requirement: FR-3

- Objective: Compute TotalAmount column on order table.

- Target Cluster Configuration:

- Cluster Name: computation-cluster
- Runtime Version: 11.3.x
- Node Type: m5.xlarge
- Driver Node: 1
- Worker Nodes: 2
- Autoscaling: Disabled
- Auto Termination: 30 minutes
- Libraries: pyspark

- Source Data Details:

- Source Type: Delta
- Location: order table

- Target Data Details:

- Table Name: order
- Storage Format: Delta

- Job Flow:

1. Load order table as DataFrame

2. Add TotalAmount = PricePerUnit * Qty

3. Write back to order Delta table

- Transformations & Logic:

- Field computation
- Type casting as required

- Error Handling & Logging:

- Capture arithmetic and null errors

TR-CORE-004

- Related Functional Requirement: FR-4

- Objective: Cleanse data of "Null"/Null literals and duplicates from customer and order tables.

- Target Cluster Configuration:

- Cluster Name: cleansing-cluster
- Runtime Version: 11.3.x
- Node Type: m5.xlarge
- Driver Node: 1
- Worker Nodes: 2
- Autoscaling: Disabled
- Auto Termination: 30 minutes
- Libraries: pyspark

- Source Data Details:
- Source Type: Delta
- Location: customer, order

- Target Data Details:
- Table Name: customer, order

- Job Flow:

1. Remove records where any critical field is "Null" (string) or null

2. Drop duplicates (all columns subset)

- Transformations & Logic:
- Record filtering
- Error Handling & Logging:
- Log records removed and reason

TR-CORE-005

- Related Functional Requirement: FR-5
- Objective: Create ordersummary target table.
- Target Cluster Configuration:
- Cluster Name: summary-creation-cluster
- Runtime Version: 11.3.x
- Node Type: m5.xlarge
- Driver Node: 1
- Worker Nodes: 2
- Autoscaling: Disabled

- Auto Termination: 30 minutes
 - Source Data Details:
 - Source Type: Delta
 - Location: as above
 - Target Data Details:
 - Table Name: ordersummary
 - Storage Format: Delta
 - Partitioning Strategy: None specified
 - Job Flow:
1. Check for table existence, create if absent
 2. Set schema per requirements
- Transformations & Logic:
 - None
 - Error Handling & Logging:
 - Log creation status

TR-CORE-006

- Related Functional Requirement: FR-6, FR-7, FR-8
- Objective: Join customer and order by CustId, applying SCD Type 2 logic for history, into ordersummary.
- Target Cluster Configuration:
- Cluster Name: scd-cluster
- Runtime Version: 11.3.x
- Node Type: r5.xlarge
- Driver Node: 1
- Worker Nodes: 4
- Autoscaling: Enabled (2-8)
- Auto Termination: 30 minutes
- Source Data Details:
- Source Type: Delta
- Location: customer, order

- Target Data Details:
- Table Name: ordersummary
- Storage Format: Delta
- Job Flow:
 1. Read cleansed customer and order tables
 2. Perform join on CustId
 3. Apply SCD Type 2 management:
 - Set inactive status, update StartDate and EndDate for changes
 4. Insert/Update into ordersummary
 - Transformations & Logic:
 - SCD Type 2 implementation
 - Join logic
 - Error Handling & Logging:
 - Log SCD changes
 - Exceptions for duplicate/inconsistent history

TR-CORE-007

- Related Functional Requirement: FR-9, FR-10
- Objective: Generate the customeraggregatespend table by aggregating spend totals per Name per Date.
- Target Cluster Configuration:
 - Cluster Name: aggregation-cluster
 - Runtime Version: 11.3.x
 - Node Type: m5.xlarge
 - Driver Node: 1
 - Worker Nodes: 2
 - Autoscaling: Disabled
 - Auto Termination: 30 minutes
- Source Data Details:
 - Source Type: Delta

- Location: ordersummary
 - Target Data Details:
 - Table Name: customeraggregatespend
 - Storage Format: Delta
 - Job Flow:
1. Read ordersummary table
 2. Group by Name and Date
 3. Aggregate sum(TotalAmount)
 4. Write final result to customeraggregatespend
- Transformations & Logic:
 - Group by / Aggregation
 - Error Handling & Logging:
 - Log aggregation anomalies

TR-CORE-008

- Related Functional Requirement: FR-11
- Objective: Orchestrate steps 1-10 using Delta Live Tables for end-to-end data pipeline.
- Target Cluster Configuration:
- Cluster Name: dlt-pipeline
- Runtime Version: 11.3.x
- Node Type: m5.xlarge
- Driver Node: 1
- Worker Nodes: 4
- Autoscaling: Enabled (2-8)
- Auto Termination: 30 minutes
- Libraries: delta, pyspark, dlt
- Source Data Details:
- Source Type: as in prior steps
- Target Data Details:
- Table Name: as in prior steps

- Job Flow:

1. Define pipeline with DLT decorators for each stage (ingestion, cleansing, transformation, SCD, aggregation)

2. Specify dependencies/order via DLT flows

3. Deploy pipeline to cluster

- Transformations & Logic:

- All operations as previous TRs, sequenced via Delta Live Tables

- Error Handling & Logging:

- Capture full DLT logs and error propagation

Sequence Diagram

